

The Dragon Hatchling

The missing link between the Transformer and models of the brain

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Reading companion · GNN / Graph-Transformer thesis exploration

★ TL;DR

A Transformer and a brain look **structurally incompatible** — dense tensors vs. a uniform, scale-free graph of local neurons. **BDH (Dragon Hatchling)** is the bridge: a language model defined as **local graph dynamics** on n neuron particles, where **attention is edge-reweighting** of synapses under a **Hebbian** rule (“neurons that fire together wire together”). Its tensor-friendly twin **BDH-GPU** trains on GPUs with **Transformer-like scaling laws** (10M–1B params, rivals GPT-2). The architecture fuses two micro-foundations: **modus-ponens reasoning** (fixed weights G) and **Hebbian fast-weights** (evolving synapse state σ). **Punchline:** BDH-GPU’s activations are **positive and sparse** ($\approx 5\%$), its trained weight graphs are **scale-free + modular** (emergent, not designed), and individual **synapses are monosemantic** — a single edge fires on “currency” across English and French. It opens a path to an **axiomatic / thermodynamic-limit** theory of reasoning models.

♦ The question

Can we build a language model that **performs** like a Transformer but admits a **brain-like** micro-interpretation — uniform, scale-free, local graph dynamics — so that its inference is **interpretable** and its scaling behavior **foreseeable**?

♦ Short answer

Yes: define the model as an **edge-reweighting kernel** on a neuron graph (attention = Hebbian synapse update). Restrict it to a **low-rank, mean-field** form $\rightarrow$ BDH-GPU, a state-space model that trains by backprop, scales in **one** neuron dimension n , and **empirically** develops scale-free modular structure and monosemantic synapses.

1 · The problem — why brains and Transformers seem unrelated

✍ My note

The motivating gap. The brain is a uniform, scale-free, **distributed** graph ($n \approx 8 \cdot 10^{10}$ neurons, $m > 10^{14}$ synapses) with **local** dynamics; the Transformer is a **centralized** tensor pipeline. Directly simulating one step of brain reasoning by a Transformer with chain-of-thought would need **billions** of CoT tokens. The paper wants a **tighter, more direct** correspondence — an architecture that is **simultaneously** a performant LM and a local graph system. This is the same “architecture-as-distributed-computing” lens as Sanford’s MPC $\leftrightarrow$ Transformer, pushed all the way to neuron-level particle dynamics.

💡 Thesis hook

Towards “Axiomatic AI” and scale-free foreseeability. The deeper agenda: a model that is **scale-free in size and time** admits a **thermodynamic limit** $\mathcal{P}_A = \lim_{n \rightarrow \infty} \mathcal{P}_A(n)$, so that small tests **extrapolate** to large deployments — a route to **PAC-like bounds for generalization of reasoning over time** (avoiding “paperclip-factory” failures of autonomous agents). This is the **safety/uncertainty** payoff and ties straight to the Hüllermeier epistemic-uncertainty reading. (*see huellermeier-2020-notes.*)

2 · The bridge — four architectures, one function

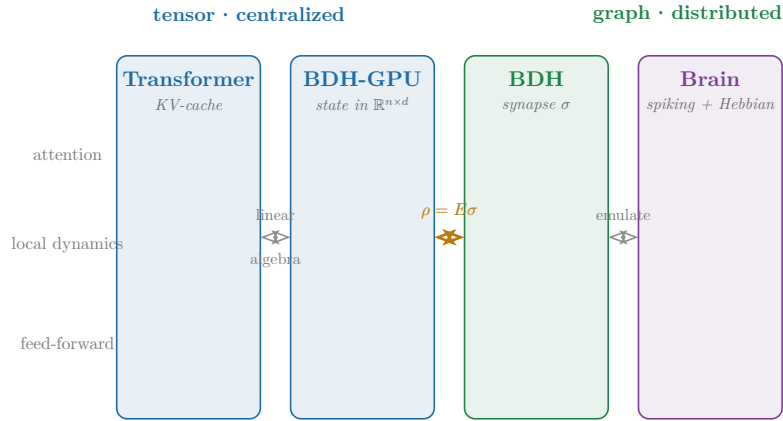


Figure 1: The central claim (paper’s Fig 1). BDH-GPU is the **keystone** of a four-way bridge: it is a **tensor** model (trains like the Transformer) yet is **formally equivalent** to BDH, a **local graph** model that a spiking Hebbian brain circuit can emulate. The same three mechanisms — attention, local dynamics, feed-forward — appear on every pier.

► Definition — BDH — a language model as local graph dynamics

n neurons (graph nodes) connected by synapses (edges). **Fixed** parameters live in neuron-neuron graphs G_x, G_y (the **ruleset** / “program”, learned by backprop); **evolving** state lives on synapse edges $\sigma(i, j)$ (**fast weights**, the working memory). Inference runs a distributed protocol of **compute** + **communicate** rounds — “communication by wire”. The system fuses two micro-foundations: **modus-ponens** inference $X(i), \sigma(i, j) \rightarrow A(j)$ and **Hebbian** reweighting $Y(i), X(j) \rightarrow \sigma(i, j)$.

📌 My note

The 1:1 parameter↔state ratio is the design key. A fast-weights system over n facts has $m = O(n^2)$ synapse state entries but comparable trainable parameters; a classic RNN (LSTM) keeps only $O(n)$ state. The Transformer’s success, the authors argue, comes from this **balanced** ratio — parameters and state both $\approx m$. BDH makes it explicit by choosing sparsity $n \ll m \ll n^2$, giving a **graph** on n nodes and m edges, where edges **triple-task** as: carrying state, carrying parameters, and **mediating communication**.

3 · The equations of reasoning

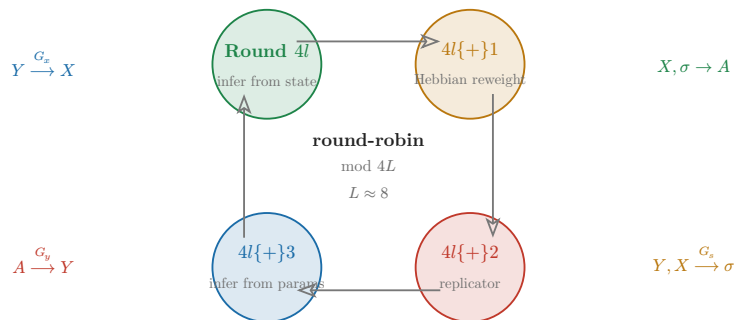


Figure 2: The “equations of reasoning” (paper’s Table 1). One BDH layer l is a cycle of four local-graph rounds: read activation from synapse **state**; **Hebbian** reweight of synapses (attention!); neuron **replicator** dynamics (evolutionary selection of active neurons); read activation from fixed **parameters**. New tokens enter every $4L$ rounds — L plays the role of Transformer depth.

► Definition — Edge-reweighting kernel

A restriction of a general chemical-reaction / replicator **interaction kernel** $q'_k := (1 - d_k)q_k + \sum_{i,j} r_{ijk}q_iq_j$ to **node-edge-node** triples $k \in \{i, j\}$ on a graph: local rules use only state on a node i , or on an edge (i, j) and its endpoints. This is **exactly** the class a spiking, Hebbian neuron graph can implement — the formal sense in which BDH is “brain-runnable”.

💡 Thesis hook

Attention as a micro-inductive bias of reasoning. Read $\sigma(i, j) > 0$ as a **positive bias toward the implication** $i \rightarrow j$ accumulated from past context. The local rule mirrors the logical axiom $(X \rightarrow (i \rightarrow j)) \rightarrow ((X \rightarrow i) \rightarrow (X \rightarrow j))$ — **modus ponens with a learned utility** on which implications to explore next. For my thesis this reframes graph attention as **programmable message passing over a learned implication graph** — a clean conceptual cousin of GNN message passing where edge weights are **dynamic, context-dependent** fast weights.

4 · BDH-GPU — the tensor-friendly twin

► Definition — BDH-GPU (n, d) state-space system, Eq. 8

Three parameter matrices: encoder $E \in \mathbb{R}^{d \times n}$ and decoders $D_x, D_y \in \mathbb{R}^{n \times d}$, shared across all L layers (Universal-Transformer style). For each token t , layer l :

$$x_{t,l} = x_{t,l-1} + (D_x \text{LN}(E y_{t,l-1}))^+,$$

$$y_{t,l} = (D_y \text{LN}(\rho_{t,l} x_{t,l}))^+ \odot x_{t,l},$$

with attention state $\rho_{t,l} \in \mathbb{R}^{n \times d}$ updated by a **rank-1 key×value** rule (linear attention). Total params = $(3 + o(1))nd$, all in E, D_x, D_y . Scales in **one** dimension n (neurons); $d \ll n$ is the low-rank **synaptic** dimension ($d = 256$, n up to 65536).

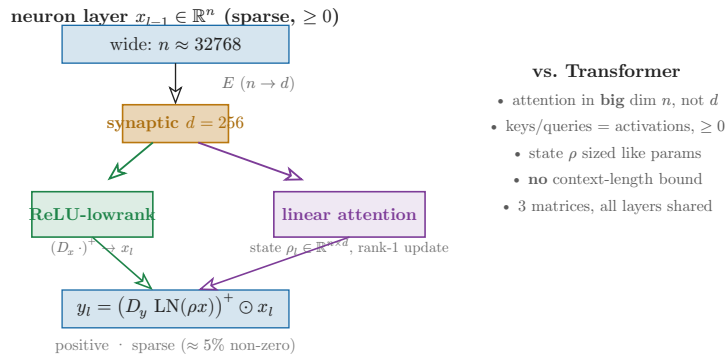


Figure 3: One BDH-GPU layer (paper’s Fig 6). Wide positive neuron activations $x \in \mathbb{R}^n$ are squeezed through the **synaptic** bottleneck $d \ll n$ by encoder E , then split into a **ReLU-lowrank** feed-forward path (D_x) and a **linear-attention** path carrying the persistent state ρ . Attention happens in the **large neuron dimension** with **positive, sparse** keys — the opposite of the Transformer’s small dense head space.

◆ Theorem — BDH $\supseteq$ BDH-GPU expressiveness (Claims 3–4, Obs. 4)

For the same asymptotic $O(nd)$ parameters, the **graph** feed-forward mechanism of BDH is **strictly more expressive** than BDH-GPU’s tensor ReLU-lowrank block (an arbitrary sparse graph $G \in \mathcal{G}^2(n, m)$ need not admit an exact low-rank factorization $G = DE$). BDH and BDH-GPU are **formally equivalent** up to LayerNorm placement, via $G_x^e - G_x^i = D_x E$, $G_y^e - G_y^i = D_y E$, $G_s = \mathbf{1}^{n \times n}$.

📌 My note

Scaling-law verdict (§4.2, Fig 7). On translation, BDH-GPU’s validation-loss-vs-size curve **tracks** GPT-2/GPT-XL across 10M–1B params, and it **learns faster per data token** (better in the scarce-

data regime). FLOPS per token $\approx O(ndL)$. It is a genuine **state-of-the-art-class** architecture, not a toy — which is what makes the brain-bridge claim weighty rather than cute.

5 · Emergent structure — scale-free, modular, monosemantic

ReLU-lowrank as denoising signal propagation (§5.3)

◆ Theorem — ReLU as a noise threshold (Claim 5)

A low-rank linear map $G = DE$ approximates a target G' only with $O(\sqrt{\log n/d})$ error in the ∞ -norm (Johnson–Lindenstrauss) — **useless** in L_2/L_1 . But adding a **negative bias + ReLU gate**, $f_{DE}(z) = (DEz)^+$, **suppresses the noise**: it can reproduce e.g. a random-walk (Markov) transition $z \rightarrow (G'z)^+$ to L_1 -error $O(\varepsilon)$. The ReLU turns a lossy low-rank channel into a faithful **sparse-positive** propagator.

✍ My note

Why modularity emerges (Claim 6, Obs. 6). A neuron j activates only when its local “F-score” $\rho = \sqrt{(|C_j| / |A|)(|C_j| / |B_j|)}$ clears a threshold — i.e. when the incoming signal is **concentrated in j ’s own cluster**. This is **precisely** communication on a graph with positive **Newman modularity**: the ReLU-lowrank net can represent in-cluster spreading on a k -block stochastic-block-model with arbitrarily small positive modularity. **Modular, scale-free structure is forced by the function**, not designed in.

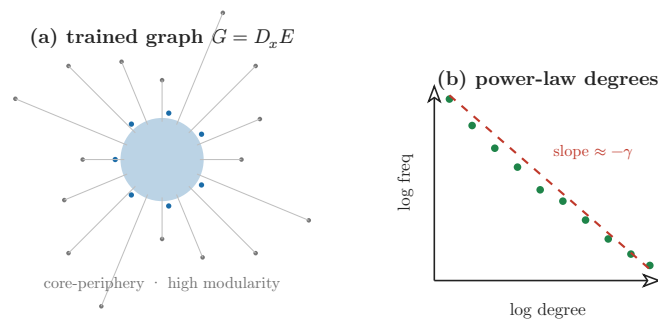


Figure 4: Emergent network structure (paper’s Figs 9–11). (a) The trained encoder-decoder product $G = D_x E$, read as a neuron-neuron graph, shows a **core-periphery, high-modularity** layout. (b) Its degree distribution is **heavy-tailed / power-law** — the litmus test of a **scale-free** system. Both arise from training alone, with no structural prior (L1-regularization was **off**).

Monosemantic synapses & sparse activation (§6.2–6.4)

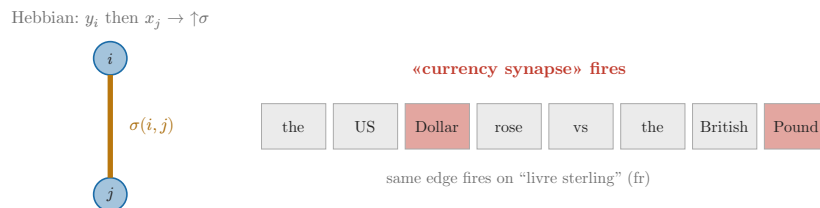


Figure 5: A monosemantic synapse (paper’s Figs 12–13). One state entry $\sigma(i, j)$ — named the “**currency synapse**” — strengthens **exactly** when a currency concept is in context, **across languages** (“Dollar”/“Pound” $\leftrightarrow$ “livre sterling”). A one-sided Mann–Whitney test confirmed the selectivity ($p < 10^{-14}$). State is readable at the level of **individual edges** — interpretability of **state**, not just weights.

✍ My note

Sparse, surprisal-driven activation. Only $\approx 5\%$ of neurons fire per token; higher layers go **quiet** once input becomes predictable (Fig 14). This is **native** adaptive computation — no separate halting network — and the reason synapse updates (and thus interpretable concept-tracking) are rare and clean. Sparsity also

enables the §7.2 experiments: **training without backprop-through-time** (only remember **which** synapse changed), and §7.1 **model merging by concatenation** in the neuron dimension.

6 · Why this matters for my thesis

💡 Thesis hook

A graph-native language model — the direct GNN/Transformer synthesis. BDH is literally a graph model whose attention is **dynamic edge-reweighting** and whose feed-forward is **local message propagation** with a ReLU threshold. This is the cleanest bridge yet between my two poles: it is a Transformer-class LM **and** a message-passing graph system. Thesis angle: position BDH as the **limit point** of the “GNN $\leftrightarrow$ Graph-Transformer” axis and ask which BDH mechanisms transfer to standard GNNs (e.g. positive-sparse activations, Hebbian fast-weight edges on a fixed graph).

💡 Thesis hook

BDH is a **distributed-computing model** — reuse the MPC lens. BDH-Normfree’s dynamics are a **mean-field broadcast** among n particles ($O(dL)$ scalars per pairwise interaction): rounds $\leftrightarrow$ layers, neurons $\leftrightarrow$ machines, $d \leftrightarrow$ local-memory/bandwidth. This is **the same dictionary** as Sanford’s MPC $\leftrightarrow$ Transformer. Thesis: place BDH on Sanford’s hierarchy — is its $O(ndL)$ -per-token, log-depth-shareable computation a **parallelizable-class** system? (see *sanford-2024-notes*, *sanford-2023-notes*.)

💡 Thesis hook

Capacity = neurons $\times$ synaptic-rank, an emergent ruler. BDH’s expressivity is set by (n, d) with state sized like parameters; emergent modularity means **effective** capacity concentrates in a scale-free core. Loukas measures GNN capacity as depth $\times$ width with a $dw = \tilde{\Omega}(n^\delta)$ floor. Thesis: compare Loukas’s **designed** depth $\times$ width capacity to BDH’s **emergent** $n \times d$ + modularity — does scale-free structure **beat** the Loukas lower bounds in practice? (see *loukas-2020-notes*.)

💡 Thesis hook

Interpretability-of-state as an epistemic-uncertainty handle. Monosemantic synapses make the **in-context state** readable per edge. Combined with the paper’s “axiomatic AI / thermodynamic-limit” goal (PAC-like bounds for reasoning over time), this is a concrete substrate for **epistemic-uncertainty** estimation on a graph model — read which concept-synapses are (un)confident. (see *huellermeier-2020-notes*.)

7 · Open questions and critique

? Open question

Equivalence vs. practice. BDH and BDH-GPU are **formally** equivalent only up to LayerNorm placement, and BDH’s sparse trainable G_s is **dropped** in BDH-GPU (set to all-ones). How much expressivity / interpretability is lost by the GPU restriction — and would a **sparse, trainable- σ** BDH (the “more expressive” version) actually train and scale? Untested.

? Open question

Emergent structure — claim strength. Scale-free / modularity is shown for 3 of 4 heads on 5 seeds via **Louvain** community detection and power-law-**looking** degree plots. Power-law fits are notoriously fragile. Is the structure **robustly** scale-free, or merely heavy-tailed? And is modularity **causal** for performance or a byproduct?

? Open question

The **axiomatic / thermodynamic-limit theory is aspirational**. The PAC-like bounds for “generalization of reasoning over time”, the existence of a limit object $\mathcal{P}_{\mathcal{A}}$, and length-generalization guarantees are **motivated** but **not proved**. Right now BDH offers a **plausible substrate** for such a theory, not the theory itself.

? Open question

Brain claim — analogy or mechanism? The bridge to spiking Hebbian circuits is at the level of **function** (attention as synapse plasticity), and explicitly **not** about longer-term learning / backprop in the brain. How much is a genuine mechanistic claim vs. a structural analogy? The §8.2 hypotheses are carefully hedged.

8 · Connections to my reading list

BDH sits at the **center** of my cluster — it is simultaneously a graph model, a Transformer-class LM, and a distributed-computing system:

- **Sanford 2024 / 2023** (Transformer $\leftrightarrow$ MPC, communication complexity): BDH-Normfree’s **mean-field broadcast** over n particles is a distributed-computing model with the **same** rounds $\leftrightarrow$ depth, memory $\leftrightarrow$ width dictionary. Natural to place BDH on the parallelizable/search hierarchy. (*see sanford-2024-notes, sanford-2023-notes*)
- **Loukas 2020** (what GNNs can’t learn, depth $\times$ width capacity): BDH is local graph dynamics with capacity $\approx n \times d$; contrast Loukas’s **designed** capacity floor with BDH’s **emergent** scale-free modular capacity. (*see loukas-2020-notes*)
- **Hüllermeier & Waegeman 2021** (aleatoric/epistemic uncertainty): BDH’s **axiomatic-AI / PAC-over-time** goal and **state interpretability** are an epistemic-uncertainty and safety angle on graph reasoning models. (*see huellermeier-2020-notes*)
- **Ye 2026** (provable learning of graph algorithms): the **learnability** counterpart — BDH **claims** SOTA learning empirically; cross-check against provable-learning results and the $\log_3 \text{diam} / \Theta(\log N)$ depth laws. (*see ye-2026-notes*)

Reading status: §1–4 (motivation, the bridge, equations of reasoning, BDH-GPU state-space) read closely · §5 (ReLU-lowrank denoising, emergent modularity/scale-free) read · §6 (linear attention, monosemanticity, sparse activation) read · §7 (model merging, no-BPTT) + §8 conclusions read · Appendices A–E skimmed · next: place BDH on Sanford’s MPC hierarchy and sketch a sparse-trainable- σ GNN variant. · *Companion generated 2026-06-30.*